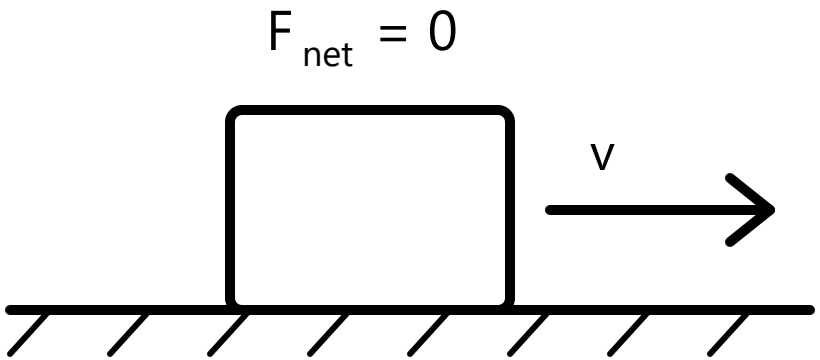
Newton's First Law: Inertia

An object at rest stays at rest, and an object in motion keeps moving at a constant velocity, unless a net force acts on it.

- Inertia is an object's resistance to changes in its motion.
- More mass means more inertia.
- Example: passengers lurch forward when a bus brakes suddenly.

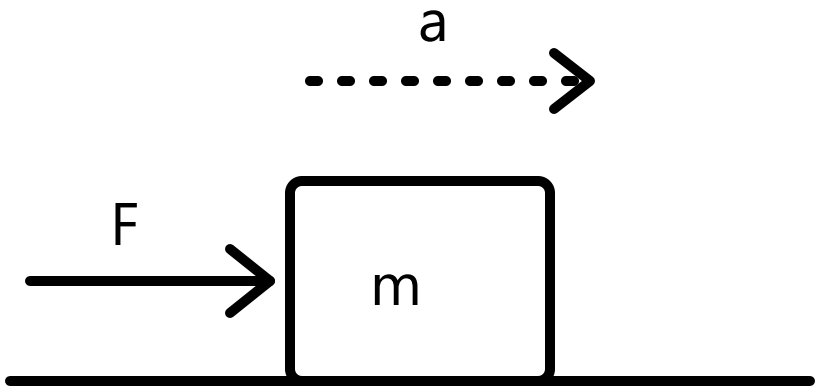


constant velocity, no net force

Newton's Second Law: F = ma

$$F = m \times a$$

- F is the net force, in newtons (N).
- m is the mass, in kilograms (kg).
- a is the acceleration, in metres per second squared (m/s²).
- 1 N = 1 kg·m/s². Double the mass with the same force and the acceleration halves.



a net force causes acceleration

Practice Problems

1. A 4 kg cart accelerates at 3 m/s². What net force acts on the cart?

2. A net force of 20 N pushes a 5 kg box across the floor. Find the box's acceleration.

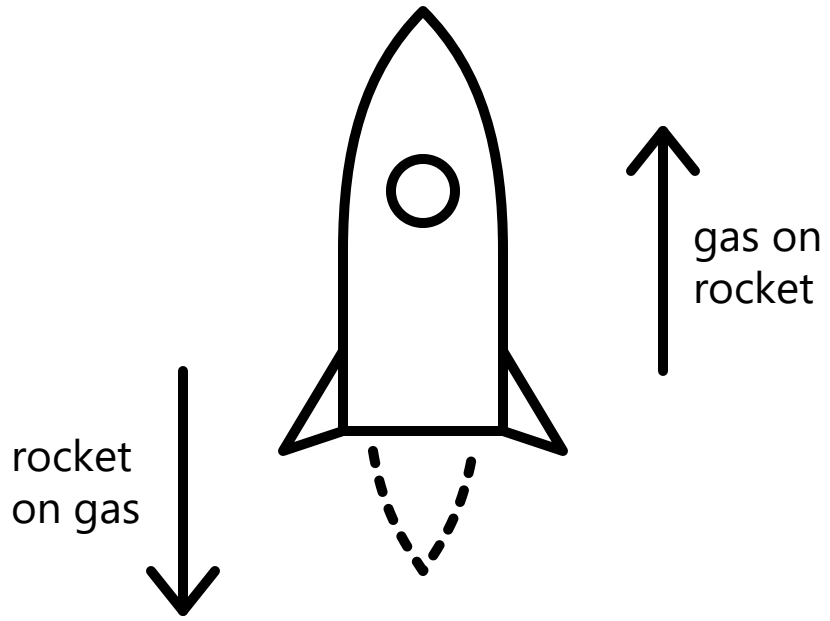
3. A 1,200 kg car speeds up at 2.5 m/s². What net force must the engine provide?

Show your working: write the formula, substitute the values, and include units.

Newton's Third Law: Action & Reaction

For every action there is an equal and opposite reaction.

- Forces always come in pairs.
- The two forces act on different objects, so they don't cancel out.
- A rocket pushes exhaust gas down; the gas pushes the rocket up.
- When you push on a wall, the wall pushes back on you.



Summary

1st law

No net force means no change in motion. Mass resists change (inertia).

2nd law

F = m × a. A bigger net force gives a bigger acceleration.

3rd law

Forces come in equal and opposite pairs that act on different objects.

Next lecture: friction and free-body diagrams.